CFA Institute

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5A team

China Southern Airlines (01055 HKEX)

Air Transport | China

Rating:	BUY
Date:	02/06/2023
Current Price:	HK$4.33
Target Price:	HK$7.642
Increase:	76.49%

2023.04	CSA
Passenger capacity input (million)	24909.01
year on year	285.87%
month-on-month	0.0722
Passenger turnover (million)	18917.33
year on year	411.55%
month-on-month	9.94%
Passenger numbers (thousands)	11319.72
year on year	403.63%
month-on-month	8.29%
passenger load factor (PLF)	75.95%
year on year	18.66%
month-on-month	1.88%

Core point

- **After the pandemic was lifted, revenue and profit grew rapidly, and the asset-liability ratio was within a reasonable range.** The company's net profit growth rate was negative during the period of the epidemic. With the release of the epidemic, the net profit growth rate reached 62.02% in the first quarter of 2023, and the revenue growth rate reached 58.61%. The company's revenue showed a good trend. The asset-liability ratio increased gradually in the past five years and reached 82.71% in the first quarter of 2023, which was within a reasonable range.
- **Capital turnover capacity has improved.** After the epidemic was lifted, the turnover rate of total assets increased slightly, and in the first quarter of 2023, the turnover rate of total assets increased 5% compared with that in 2022. Among them, the inventory turnover rate was 69.67% in 2022, maintaining a high level and higher than the equivalent level of the industry. Accounts receivable decreased by 8.22% year on year in 2022, and the receivable ratio began to be higher than the peer median in 2022, but remained below 15%.
- **Demand side travel improved, China Southern Airlines reduced losses in the first quarter of 2020, and the subsequent recovery is expected.** The net profit margin in the first quarter of 2023 was -9.83%, which was 0.78% lower than that in 2022, mainly due to the improvement of operating profit margin. Operating revenue was 34.055 billion yuan, an increase of 12.58 billion yuan year-on-year, mainly benefiting from the increase in the number of air travelers. Operating costs were 33.294 billion yuan, up 33.6 percent year on year.
- **Supply side capacity improvement, passenger load factor recovery of the three leading airlines**. In April 2023, China Southern Airlines passenger capacity input increased by 285.87% year on year to 24.909 billion passenger-kilometers, passenger turnover increased by 411.55% year on year to 18.917 billion passenger-kilometers, passenger transport increased by 403.63% year on year to 11.32 million person-times, load factor was 75.95%. The passenger load factor of China Southern Airlines recovered to that of the same period in 19 year 91.5 percent, leading among the three airlines.
- **Investment logic:** With the gradual recovery of domestic and international routes, China Southern Airlines has a relatively good domestic traffic situation, overseas recovery in Southeast Asia also has more routes. Based on the company's

valuation model, we believe the stock should be priced at 6.90 yuan by the end of 2023。Combined with its relatively healthy financial position, the company has great potential for operating revenue. We rate China Southern Airlines Limited a "buy" rating.

Company Profile

China Southern Airlines Company Limited, established on March 25, 1995, is directly affiliated with China Southern Air Holding Limited Company, specializing in air transportation services. The main business is to provide domestic, Hong Kong, Macao and Taiwan regions and international air passenger, cargo and postal services. China Southern Airlines Co., Ltd. is an AH Shares Dual Listing Company with a total share capital of 18.121 billion. A-shares account for 74.37% of the total share capital and H-shares account for 25.63% of the total share capital. Such a share issuance model largely meets the company's need to maximize financing.

Subsidiary

China Southern Airlines Co., Ltd. has 18 holding subsidiaries, 12 joint ventures and shareholding companies, and 3 joint ventures. The business involves air transportation, aviation technology, data communication, aircraft maintenance and other fields, basically covering the upstream, midstream and downstream of the air transportation industry. The layout is broad, the business scope is broadened, and the foundation is laid for the company's future development.

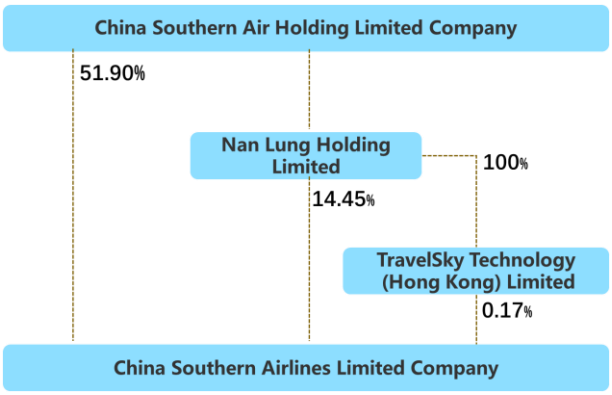


Figure 1 :Shareholding structure

Capital Structure

From the data point of view, the capital structure of China Southern Airlines Co., Ltd. is relatively concentrated. The largest shareholder, China Southern Air Holding Limited Company, holds 51.9% of the shares, followed by Nan Lung Holding Limited, which holds 14.41%, and the remaining shareholders hold less than 10% of the shares. Besides, Nan Lung Holding Limited is a wholly-owned subsidiary of China Southern Air Holding Limited Company in Hong Kong. As a typical group holding company, China Southern Airlines Co., Ltd. can accept the unified command and deployment of the parent company to reduce costs and absorb more external funds.

Strategic direction

China Southern Airlines Co., Ltd. has always been committed to building a world-class air transport enterprise with global competitiveness and enhancing global competitiveness.

Corporate Governance

Governance Structure

China Southern Airlines Co., Ltd. has established the shareholders' meeting, board of directors, and board of supervisors. And the Board of the Company has put in place an Aviation Safety Committee, a Strategic and Investment Committee, an Audit and Risk Management Committee, a Nomination Committee and a

Remuneration and Assessment Committee. The number of independent non-executive directors accounted for more than a half in Strategic and Investment Committee, Audit and Risk Management Committee, Nomination Committee and Remuneration and Assessment Committee.

The company has a complete corporate governance structure and clearly defines manager responsibilities, which provides an important guarantee for the company's internal stability and long-term development.

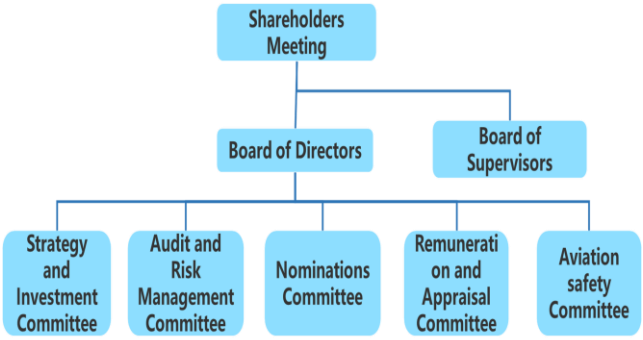


Figure 2: Corporate structure

Composition of the Board of Directors

The company's board of directors consists of seven directors, including three executive directors and four independent non-executive directors. Among them, the 3 executive directors are from the parent company, representing the interests of shareholders; the 4 independent non-executive directors are high-end talents involved in engineering technology, management, finance, news and communication and other fields.

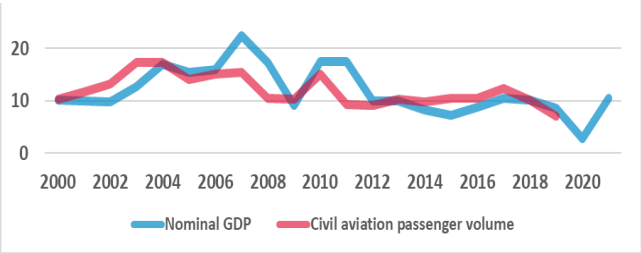
An effective board needs to have independence experience and resources necessary to protect the shareholders' long-term interests. The composition of personnel in multiple fields meets the needs of China Southern Airlines' current and future development. It can make informed decisions about the firm's future, ensuring that the assets of the company are used efficiently.

Industry Analysis

Basic overview of the industry

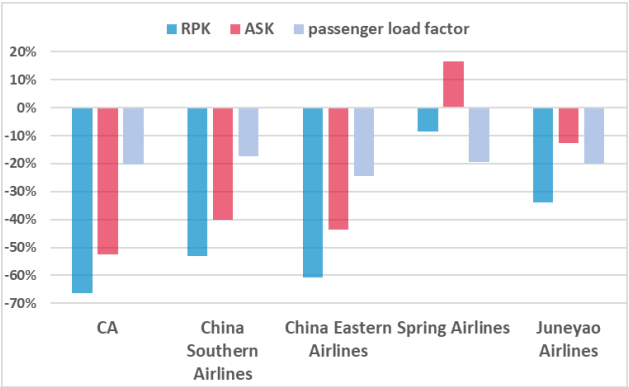
China Southern Airlines Corporation Limited belongs to the civil aviation transport industry, which is a traditional service industry with heavy assets. As the basic industry of transportation, civil aviation transport industry has the basic characteristics of infrastructure industry such as high investment, high yield and high efficiency. For China's aviation industry, the upstream industry is mainly jet fuel suppliers, aircraft manufacturers and airports, while the downstream industry is passengers and enterprises or individuals who need to carry out air cargo transportation. China's civil aviation transport has developed for more than thirty years, after three government-led mergers and reorganizations, and the deep integration of aviation industry is the main trend of the development of China's civil aviation transport industry. The civil aviation transport industry has the basic characteristics of high periodicity, high policy control, capital intensive and technology intensive. In addition, the supply and price of aviation fuel, the change of exchange rate and interest rate, and the degree of government subsidy income are also important factors affecting the development of the industry, and they are difficult to avoid.

Figure 3: Comparison of China's passenger traffic growth rate and nominal GDP growth rate



Source:Yidu Data

Figure 4: Changes in sales of major listed airlines from January 2022 to January 2019



Source:Yidu Data、Company announcements

Macro environment

The growth of downstream passenger traffic demand of civil aviation industry is consistent with the growth of domestic GDP, and it is a long-term trend that industry demand is closely related to

residents' disposable income. When the economy is in the stage of steady growth, business activities are correspondingly more active, and the travel demand of business travelers increases substantially. However, due to the impact of COVID-19 in recent years, China's GDP growth has slowed down significantly. The demand side of the civil aviation transportation industry has been greatly impacted, and the realization of the impact is still delayed. During the epidemic period, affected by the international situation, the epidemic prevention policies of all countries showed a negative attitude towards cross-border air travel. Airlines successively reduced the capacity of international routes to match the low shipping demand, and the industry transport volume suffered a cliff drop year-on-year. On the whole, several major operating indicators of the airline industry, such as RPK, ASK, all had negative growth to varying degrees. However, as the domestic epidemic situation gradually stabilized, the civil aviation transport situation gradually recovered in 2023, and the loss narrowed. With the May Day holiday and the peak of summer travel, the demand of civil aviation transport industry has increasing trend, the industry is expected to return to profitability and is expected to continue to recover, the industry growth period is longer.

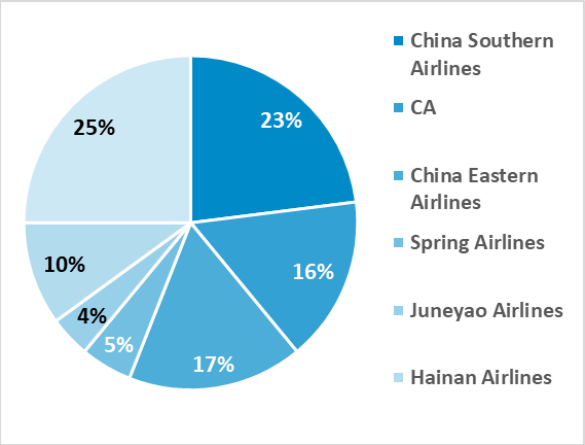
On the supply side, changes in the supply pattern of the civil aviation industry are often caused by world events. Affected by the OPEC production cut, the Russia-Ukraine conflict and other events, the price of oil has been pushed up significantly. As the downstream product of oil, the price of aviation fuel resonates with the world oil price. It is difficult to make up the supply gap quickly, which will lead to a sharp rise in the cost of domestic jet fuel. Due to the limited hedging policies and countermeasures, Air China is highly sensitive to oil price fluctuations, which will further squeeze the airline's profit space.

Table 1: Fuel costs of three major airlines

Airline	Fuel cost in the first half of 2021(billion yuan)	1% increase in oil price corresponds to a rise in aviation fuel cost (billion yuan)
CA	123.36	0.99
CEA	99.15	0.98
CSA	97.73	1.23

Source: International aerial navigation

Figure 5: Ratio of caliber turnover volume of China's civil aviation industry in 2021



Source:Yidu Data、Civil Aviation Administration

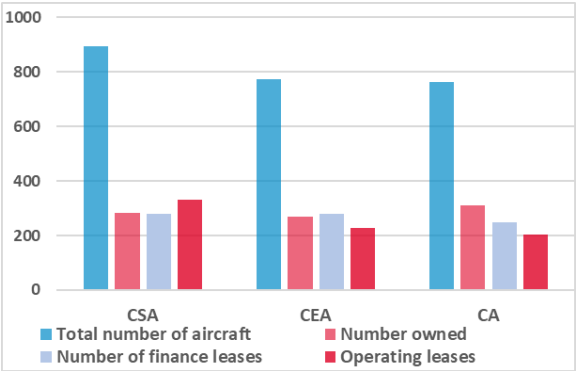
Competitive advantage in the industry

1.Significant advantages of economies of scale and perfect network system

The total transport turnover of China Southern Airlines has ranked first in China in recent years. It has the largest fleet, with a total of 894 airliners, featuring various types and excellent fleet performance. It has the densest airline network, forming a developed airline network that covers China intensively, fully radiates Asia, and effectively connects Europe, America, Australia and Africa. With the largest annual passenger transport volume, it is the first domestic airline with a transport volume of more than 100 million. In addition, the company has also set up 7 regional marketing centers, and has 23 domestic sales departments and 55 overseas sales departments across all continents, forming a point-and-point combination of branches, holding companies, regional marketing centers and sales departments.

2.High standard brand service level

Figure 6: Aircraft holdings of major civil aviation companies in China



Source:China Business Information Network

The company carries out large service construction, to build the international first-class service brand while constantly improving the quality of service, to "friendly and fine" as the service positioning, break the barriers between various systems and departments, to achieve the whole chain, systematic, integrated service management, has launched a series of new service products, service measures, the company's service quality and flight on-time rate continue to improve, to achieve industry leadership, The company has won the title of "National Customer Satisfaction Benchmarking Enterprise" for many times.

3.Leading industry's informatization technology level

At present, the company has a large team of information-based talents. The company pays attention to relevant technology research and development, and has auxiliary support of relevant systems in the fields of passenger transport marketing, operation control, aviation safety and public platform. It has initially realized the technical framework of "cloud platform + double medium platform", and created a platform of enterprise-level digital empowerment and capability reuse, which supports the strategic transformation and comprehensive development of the company's business.

4.Obvious location advantage

The company continues to strengthen the integration of the Guangdong-Hong Kong-Macao Greater Bay Area and promote the connectivity between the Greater Bay Area City cluster and major global city clusters. In addition, the company focuses on the construction of two comprehensive international hubs in Guangzhou and Beijing, all of which transfer to Daxing Airport in Beijing, and become the airline with the highest market share in Daxing Airport. By taking advantage of resource endowment, comprehensively promoting the strategic layout of "dual hub", improving the system mechanism and supporting resources, the company has gradually formed a new development pattern of "North and South echo and fly side by side".

Environmental analysis based on Porter's Five Forces model

1.Competitive rivalry

China Southern is the largest player in the industry and is able to exploit economies of scale. Affected by the epidemic, some small and medium-sized airlines with poor performance in China have been taken over by local SASAC. At present, SASAC is leading the industry integration with the three airlines as the core. As one of the three airlines, China Southern enjoys certain advantages. In addition, there are some inter-provincial differences in the market competition among the three airlines, with different proportions of domestic and international business. In general, the current competitive environment of the industry is relatively stable, the number of existing competitors is not large, the degree of competition is relatively low.

In the industrial chain, China Southern Airlines belongs to the midstream service provider, and its upstream business is equipment provider and aviation and airport operation Enterprise, its

downstream is mostly for business travel or private travel service objects.

2.Bargaining power of suppliers

In terms of production, upstream of the civil aviation transport industry is the aircraft manufacturing industry, which is dominated by Boeing and Airbus. Domestically produced aircraft have yet to occupy a large scale in the market. In addition, the upstream companies have a serious backlog of orders, and the aviation company has no independent production capacity, which is deeply restricted by the capacity of the aircraft manufacturing industry.

In terms of energy supply, the aviation company is extremely dependent on suppliers of aviation fuel and materials and suppliers of airport facilities. China Aviation oil Company and aviation supplies Company are the administrative departments of the state with monopoly management rights, and the aviation company is vulnerable to the influence of administrative policies.

Generally speaking, the airline company of China Southern Airlines has poor bargaining power and lacks the initiative in bargaining price.

3.Bargaining power of customers

At present, China's aviation market has been transformed from a seller's market to a buyer's market. Although individual passengers lack the bargaining power with airlines, there is competition among airlines. Some travel agencies and agents have a large number of passengers, which improves the bargaining power of buyers. A small number of agents also use the internal information of China Southern Airlines to further bargain with the sellers. The airline also lacks bargaining power.

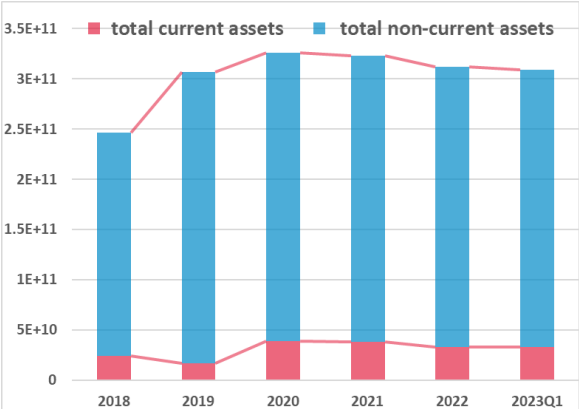
4.Threat of new entrants

The air transport industry is heavily regulated, asset-heavy, highly indebted and has a long capital recovery cycle. First of all, there is a relatively strict policy environment in China, and stable supervision helps China Southern maintain its first-mover advantage, fully enjoy the dividends of the pattern, and form an industry pattern of constant strength of the strong. Secondly, the civil aviation industry requires continuous heavy investment in assets. It is a typical capital-intensive industry with high cost of numerous flight and management equipment, difficulties in personnel training, and great sunk costs. Finally, aviation resources are scarce. Restricted by airspace, airports and routes are both scarce resources, which is not conducive to the competition of too many airlines. China Southern Airlines, as the leading airline, can squeeze out small airlines at a lower cost. Therefore, the air transport industry has very high industry barriers, and the new entrants pose less threat to China Southern Airlines.

5.Threat of substitute

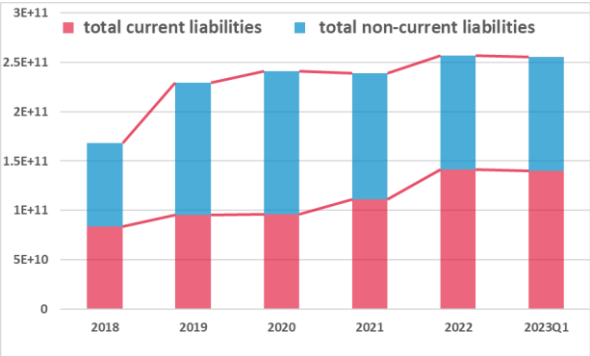
As an industry of infrastructure construction, civil aviation transportation plays an irreplaceable role and position in the national comprehensive transportation system. Moreover, it occupies a special and important position in long-distance high-speed passenger transportation and international passenger transportation, with little

Figure 7: Asset change and proportion



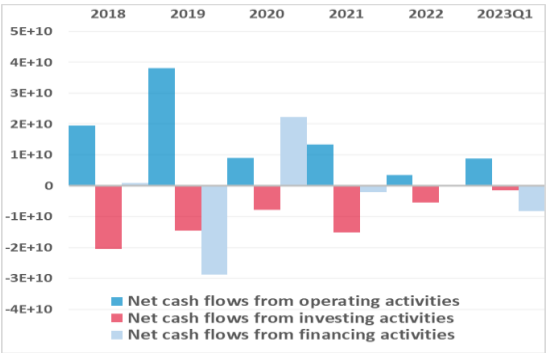
Source:CSMAR

Figure 8: Change and proportion of liabilities



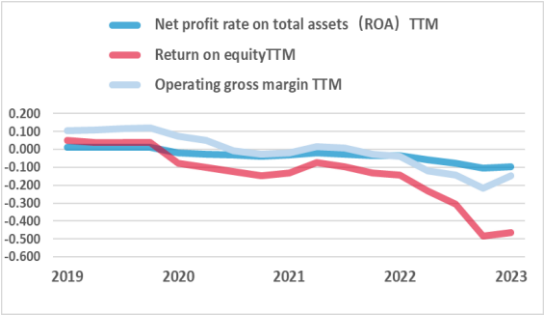
Source:CSMAR

Figure 9: Cash flow change



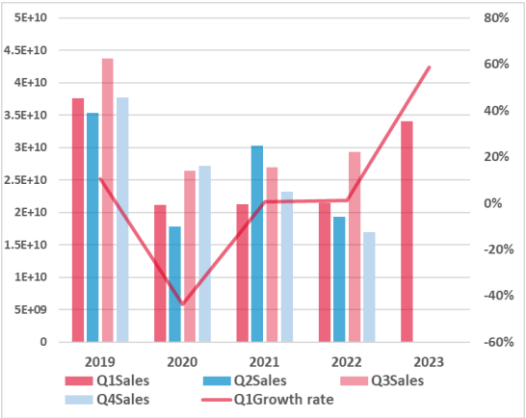
Source:CSMAR

Figure 10: Annual ROA、ROE、gross margin



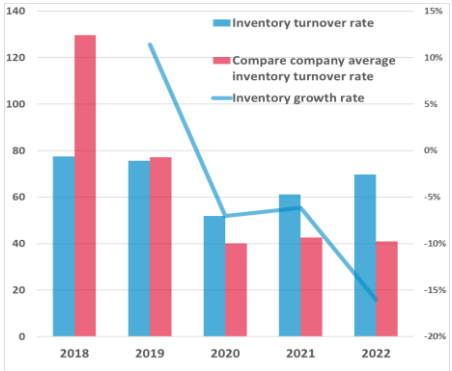
Source:CSMAR

Figure 11: Quarterly sales and growth rate



Source:CSMAR

Figure 12: Annual inventory turnover and survival rate



Source:CSMAR

substitutability. However, for the market segment of the transport industry, China has built the world's largest high-speed rail network system. The sustainable development of high-speed rail will divert some passengers and bring negative effects to the air transport industry, so there is certain substitutability.

Financial analysis Asset allocation and management

Asset allocation and management

1.Asset Analysis

The asset scale of China Southern Airlines is stable on the whole. From the perspective of asset composition, the assets of the company are mainly non-current assets, accounting for 89.37% of the total assets in the first quarter of 2023. The internal changes of assets are mainly caused by the purchase of aircraft offices

2.Liability analysis

Debt is on the rise as a whole. From the perspective of debt structure, the company's liabilities are mainly current liabilities, current liabilities account for 54.91% of the total liabilities in the same period, and short-term loans account for the largest proportion of 30.84%. At the end of 2022, the company's short-term loans increased by 17.094 billion yuan compared with the end of the previous year, an increase of 71.32%. The changes of short-term and long-term borrowings are mainly caused by the changes of bank loans.

3.Cash flow management

Net cash flow from operating activities in January-March 2023 increased by 53.43 billion yuan compared with the beginning of the year, mainly due to a substantial increase in sales revenue. Investment activities mainly increased the purchase of aircraft, issuers have smooth financing channels and strong financing ability, the company's cash expenditure in financing activities is mainly the cash paid for debt repayment.

Financial index analysis

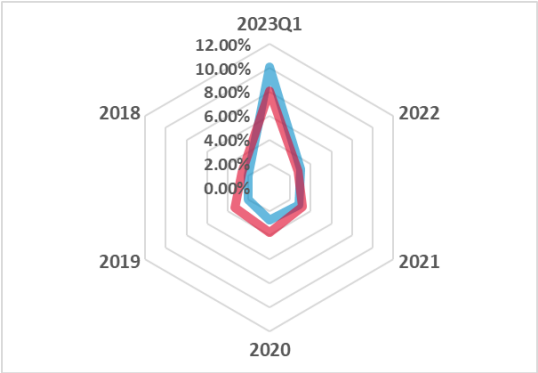
1.Profit ability

Affected by the epidemic, China Southern's return on assets、equity and gross profit rate have shown a downward trend since 2020. In the first quarter of 2023, the return on equity and gross profit rate have picked up slightly, increasing by 1.95% and 6.95% respectively compared with the end of the year. From perspective of quarterly (year-on-year) sales growth of China Southern Airlines, the sales volume in the first quarter of 2023 was 34055000000, with year-on-year growth 58.61% , showing a good turnaround trend, with the arrival of the peak season of aviation, the profitability of Southern Airlines will be further improved.

2.Operation capacity

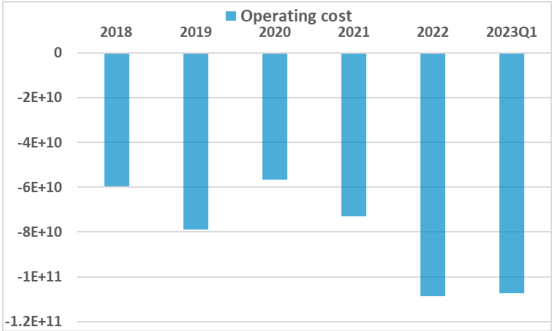
The total asset turnover ratio of China Southern Airlines in 2022 is 0.274, showing a downward trend year by year. The inventory turnover in 2022 was 69.67, remaining high and above peer parity.

Figure 13: Comparison of accounts receivable as a percentage of revenue



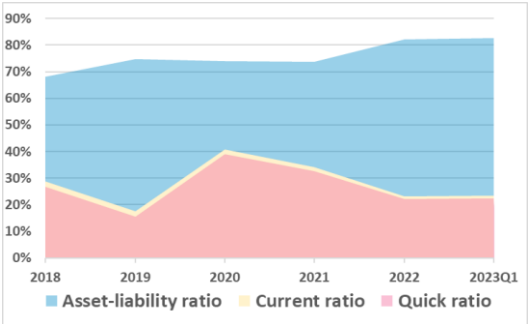
Source:CSMAR

Figure 14: Asset-liability ratio, current ratio, quick ratio changes



Source:CSMAR

Figure 15: Changes in working capital



Source:CSMAR

Accounts receivable amount decreased by 8.22% year on year 2022. The receivable ratio began to be above the peer median in 2022, but remained below 15% and within reasonable limits. In the first quarter of 2023, with the gradual relief of the epidemic, the turnover rate was correspondingly increased.

3.Debt paying ability

China Southern's current ratio and quick ratio in the first quarter of 2023 were 0.23 and 0.22 respectively, both lower than the industry median 1.15,1.12. The company's current ratio and quick ratio are low, its working capital is negative from 2018 to 2023Q1, and its asset composition is mainly non-current assets, which is in line with the characteristics of the industry the company is in. Monetary capital accounts for a relatively high proportion of the company's current assets, and the company has strong liquidity. In the first quarter of 2023, the operating revenue increased by 58.61% compared with the same period last year, and the short-term solvency was strengthened. The asset-liability ratio increased to 82.71% in the first quarter of 2023. The asset-liability ratio was generally on the rise year by year, and the asset-liability ratio remained in a reasonable range.

Generally speaking, the company implements relatively sound financial policies, and the debt-to-capital ratio is at a reasonable level in the industry. The strong cash earning ability of the company provides a stable and reliable source for the repayment of debts, and the overall solvency is strong.

4.Growth ability

Due to the impact of the epidemic, the growth rate of China Southern's total assets showed a significant decline in 2019, and the growth rate remained at -3.46% in the first quarter of 2023. In 2022, both operating revenue and net profit grew negatively, and in the first quarter of 2023, the growth rate recovered to a positive growth rate, which was 58.61%, and the net profit margin was 62.02%. With the improvement of the epidemic situation and the recovery of tourism, China Southern Airlines, as the only choice of travel, has a strong ability to grow and develop.

DuPont analysis

From the conclusion of DuPont analysis (Appendix 1), the total assets turnover ratio is slightly higher than the industry average, which improves China Southern's return on equity, among which the fixed assets turnover ratio and current assets turnover ratio are both higher than the industry average. From the perspective of ROE decomposition, the three indexes are all better than the average of the peer group. Under the same macro environment, they show good profitability, operating ability and debt paying ability.

Risk warning

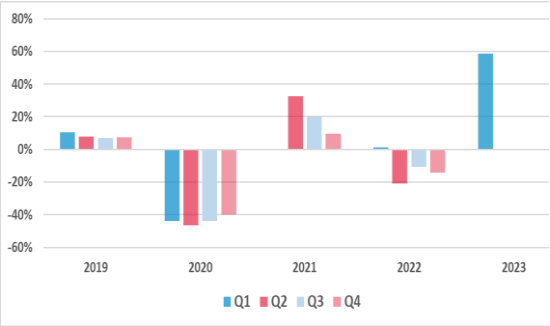
Market competitive pressure

First of all, with the gradual saturation of the aviation market, the competition among airlines is becoming increasingly fierce, and there are more and more means such as price war and preferential policies.

Figure 16: Change in net profit growth rate

Source:CSMAR

Figure 17: Changes in the growth rate of operating revenue



Source:CSMAR

Table 2

Calculate the data needed for Kd, the cost of debt	
SD short-term debt	121842 (million RMB)
LD long-term debt	105996 (million RMB)
TD total debt	227838 (million RMB)
TN short-term interest rat	2.10%
TB long-term interest rate	2.64%
TR effective tax rate	16.50%
AF debt adjustment coeffi	1.04

Source: HKEX、NBS

Table 3

Calculate the data needed for Ke, the cost of equity	
Rf risk-free rate	2.46%
β beta	1.02
Rm The expected yield of the market	-17.70%

Source: HKEX、NBS、Flush

The intensified market competition may bring uncertain risks to the development of the company. Secondly, affected by the epidemic, since 2020, the operating gross margin of China Southern Airlines has shown a downward trend, indicating that the competition in the aviation industry is constantly intensifying. As a company with a large market share, China Southern Airlines must face the challenges of other airlines, while maintaining its own market position.

Risk of large capital expenditure scale

The internal changes of assets of China Southern Airlines are mainly caused by the purchase of aircraft. The operating costs of the airline mainly include fuel costs, labor costs, maintenance costs, etc, and the fixed operating costs of these aircraft are constantly rising. Therefore, it is necessary to constantly improve the operating situation to prevent a sharp decline in operating profit, financial difficulties and other problems.

Risk of interest rate fluctuations

The asset-liability ratio of China Southern Airlines increased to 82.71% in the first quarter of 2023, and the asset-liability ratio is generally rising year by year. Therefore, the interest rate fluctuation caused by the tight change of market funds has a great impact on the company's financial expenses, thus affecting the company's operating performance.

Principal-agent

Recently, the relevant airlines appear in the process of service language discrimination, bad service attitude and other phenomena, this phenomenon is bound to affect the main business of the relevant airlines. As a service industry, it is necessary to provide relevant training for service personnel and supervision personnel, strengthen agency service management, improve the corresponding work process and the external agency service guarantee level, and patiently serve every customer.

Company valuation

China Southern Airlines Co.Ltd is a civil aviation transportation company, which belongs to the traditional service industry that focuses on assets. Due to its ROE<k, relatively stable cash flow and other characteristics, we use two valuation methods: absolute valuation (DCF model) and relative valuation (P/E, P/B) to estimate the intrinsic value of the company. Finally,considering the advantages and disadvantages of different valuation methods, from a comprehensive perspective, we respectively give a certain weight to the valuation results obtained by the two methods and then add them together to get the final target price.

Absolute valuation (DCF model)

1.Calculating WACC

(1) Calculate the Kd ： Kd=3.58%

(2).Calculate the Ke： Ke=-18.10

(3).Calculate WACC： WACC=0.59%

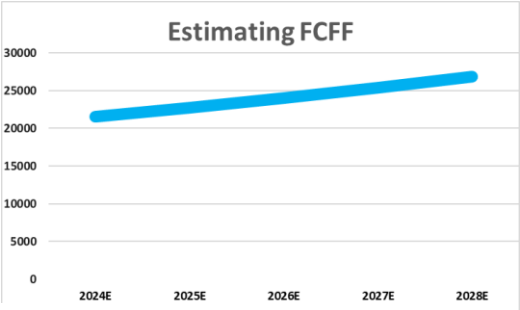
2.Estimating FCFFt

Table 4

Calculate the data required for the weighted average cost WACC	
Ve equity capital value	29283 (million RMB)
Vd debt capital value	227838 (million RMB)
V Total capital value	257121 (million RMB)
Ke equity cost	-18.10%
Kd debt cost	3.58%
TR effective tax rate	16.50%

Source: HKEX、Wind

Figure 18: Estimating FCFF



Source: HKEX、Wind

Table 5

Stock code	Comparable Companies	P/E	P/B
601111	Air China	loss	4
600115	ChinaEastern Airlines	loss	4.13

Source: Flush

We have estimated FCFFt for the next five years based on China Southern Airlines Limited's free cash flow and growth rate over the past five years, as follows (in millions of RMB).

3.Calculation of Stock Price (using DCF Model)

PV=118382.97 (millions of RMB)

From HKEX, the number of shares is 17205 (million shares)

Share price =PV/ number of shares =6.88 (yuan/share)

Relative valuation (P/E and P/B models)

1.Collection of comparable company data

We use both price-to-earnings ratio (P/E) and price-to-book ratio (P/B) to value the underlying companies. We select two companies with similar market value and similar business types as comparable companies, namely Air China and China Eastern Airlines.

2.Analysis of weight allocation

As China Southern Airlines Co., Ltd. and two comparable companies have been loss-making for a long period of time in the past five years, we abandoned the P/E model and used the P/B model entirely to estimate the share price. Since there is no significant difference in the referrability of the two comparable companies to China Southern Airlines Co., LTD., a weight of 50% is assigned to each.

3.Calculation of share Price (using P/B model)

A query from HKEX HKEX shows a net asset value per share of 1.70. The final price is 6.91 yuan per share.

Take the valuation results into account

In view of the impact of macro environment, the whole H-share market is in a low valuation condition. In order to better reflect the reality, we give the weights of 30% and 70% to the absolute valuation target price and relative valuation target price respectively. Finally, the target price of China Southern Airlines Co., Ltd. is 6.90 (yuan/share).

Investment logic

Based on industry development

With the gradual recovery of domestic and international routes, China Southern Airlines has a relatively good domestic traffic situation, overseas recovery in Southeast Asia also has more routes. In the long run, China's economy will develop at a stable growth rate, and the demand for air transport services will continue to rise. The industry development will maintain the trend of recovery, with promising potential. At present, we give China Southern Airlines in the air transport industry rating as "promising".

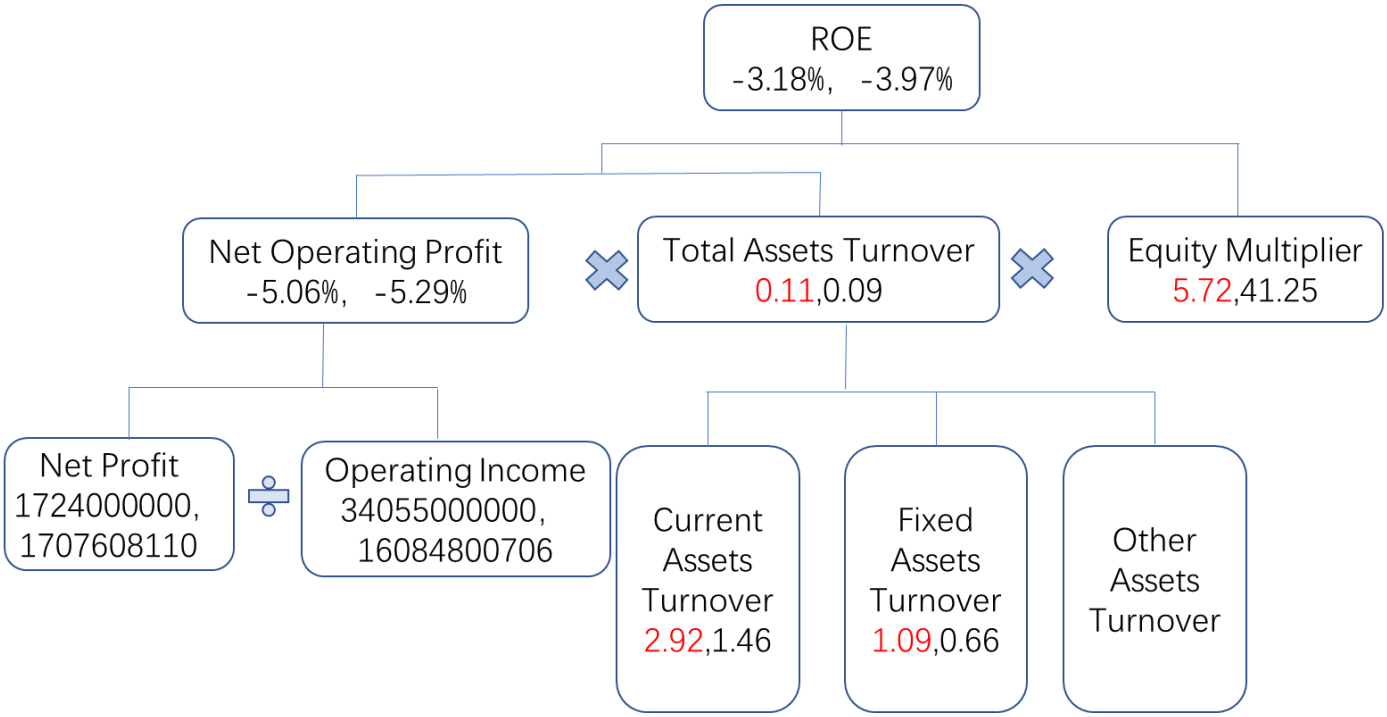
Based on the development of the airline market

The current share price of China Southern Airlines Co. Ltd. is 4.57 yuan. Based on the company's valuation model, we believe the stock should be priced at 6.90 yuan by the end of 2023, making the stock undervalued. Combined with its relatively healthy financial position, the company has great potential for operating revenue. In summary, we rate China Southern Airlines Limited a "buy" rating。

Appendix 1: Financial Ratio

Rate	2018	2019	2020	2021	2022	2023Q1
Currency						
Current ratio	29.00%	17.53%	40.74%	34.12%	23.19%	23.42%
Quick ratio	26.82%	15.55%	38.91%	32.63%	22.21%	22.32%
Financial leverage						
Asset-liability ratio	68.22%	74.81%	73.92%	73.85%	82.27%	82.71%
Operational capability						
Accounts receivable turnover rate	51.30	50.40	32.25	37.38	31.37	30.40
Inventory turnover	77.45	75.54	51.96	61.10	69.67	68.34
Turnover of fixed assets	0.87	1.21	1.09	1.15	0.96	1.09
Total asset turnover	61.78%	55.78%	29.26%	31.32%	27.42%	31.70%
Profitability						
Gross profit margin	10.45%	12.09%	-2.53%	-2.54%	-21.60%	-14.64%
Operating margin	2.79%	2.07%	-16.90%	-14.07%	-36.99%	-27.86%
Net profit margin	1.49%	1.12%	-3.74%	-3.39%	-10.61%	-9.83%
Return on equity	4.92%	3.99%	-14.60%	-13.02%	-48.36%	-46.41%
Return on assets	4.13%	4.17%	-3.85%	-3.04%	-7.02%	-0.19%
Developing abilities						
Revenue growth rate	12.66%	7.45%	-40.02%	9.81%	-14.35%	58.61%
Net profit growth rate	-49.42%	-10.45%	-481.91%	6.84%	-206.04%	62.02%
Total profit growth rate	12.97%	24.32%	6.35%	-0.97%	-3.39%	-3.46%

Appendix 2: DuPont Analysis



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